for the conditions to be satisfied: the maximum size is 40 GB (`"max_size": "40gb"`) or the age is older than six days (`"max_age": "6d"`). Once one of these conditions is met, the index transitions from the hot phase to the warm phase.

- *Warm phase*—When the index enters the warm phase, it stays there for a week (`"min_age": "7d"`) before any of its actions are implemented. After the seventh day, the index is shrunk to one node (`"number_of_shards": 1`), and then the index is deleted.
- *Delete phase*—The index stays in this phase for 30 days (`"min_age": "30d"`). After this time lapses, the index is deleted permanently. Be wary of this stage, because the delete operation is irreversible! My advice is to make a backup of the data before you delete the data permanently.

It's time to wrap up. In this chapter, we learned a lot about indexing operations and ILM. In the next chapter, we go over text analysis in detail, so stay tuned.

Summary

- Elasticsearch exposes index APIs to create, read, delete, and update indexes.
- Every index has three sets of configurations: aliases, settings, and mappings.
- Indexes can be created implicitly or explicitly:
  - Implicit creation kicks in when an index doesn't exist and a document is indexed for the first time. Default configurations (such as one replica and one shard) are applied on an index that's created implicitly.
  - Explicit creation occurs when we instantiate indexes with a custom set of configurations using the index API.
- An index template lets us create indexes with predetermined configuration settings that, based on a matching name, are applied during index creation.
- An index can be resized using a shrinking or splitting mechanism. Shrinking reduces the number of shards, while splitting adds more primary shards.
- An index can be conditionally rolled over as required.
- Index lifecycle management (ILM) helps transition indexes between these lifecycle phases: hot, warm, cold, frozen, and delete. In the hot phase, the index is fully operational and open for searching and indexing; but in the warm and cold phases, the index is read-only.